SAM4 HEALTH & Edge Analytics

SAM4 Health is now equipped with Edge Analytics

SAM4 Health is an AI-driven asset health monitoring system. It allows global industries to continuously monitor critical assets located anywhere from ATEX zones to submerged environments in order to detect developing faults and prevent unplanned downtime. It is now equipped with edge analytics capabilities to enable companies to profit from the same benefits the system always offered, but now in reduced bandwidth settings. This means access to advanced analytics on machine health, diagnostics on developing failures up to months in advance, and automated and expert maintenance recommendations. All this insight comes continuously, in real time and with a very high level of accuracy. For many industries this will be the first time that an asset health monitoring system with such advanced capabilities will be available for employing in low-bandwidth settings.

PRODUCT FEATURE EXPLAINER

Samotics

The challenge of monitoring critical assets in remote, low-bandwidth environments

Continuous real-time monitoring is not an easy task to accomplish as it generates massive amounts of data that need to be stored, processed and analyzed. For many organizations with critical assets located in remote areas with limited and unreliable network connectivity, it can become a big hurdle to implement asset health monitoring systems. For example, industries that have:

- data transfer issues due to connectivity
- remote or offshore locations with only a satellite link

Some of these industries will also face other critical issues around asset maintenance. Downtime and maintenance could result in significant financial costs, cause safety issues and have environmental consequences. There could be limited on-site storage capacity for spare parts. Some industries could additionally be facing a future shortfall of qualified and experienced maintenance personnel. At the same time there is the need to keep operating costs down without increasing risk. This puts additional pressure on maintenance planning. This is where edge analytics capabilities built into SAM4 Health solve these problems.

How Edge Analytics enables SAM4's capabilities in low-bandwidth settings

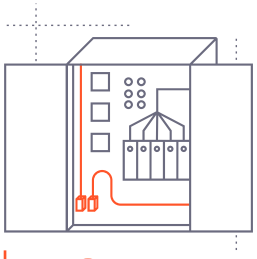
The edge analytics functionality has significantly reduced the amount of bandwidth SAM4 Health requires to perform the same high level of advanced AI-based analytics on machine performance. SAM4 Health failure detection algorithms have now been updated to use a set of failure indicators developed from the raw, high-frequency electrical data rather than using the raw data itself. This allows the SAM4 platform to perform real-time failure diagnosis on a greatly reduced data set.

The algorithms developed to calculate these failure indicators have been transferred to the SAM4 gateway, which is located at our customers' sites. The gateway now pre-processes the raw electrical data to reduce it to the associated failure indicators before transferring it to the SAM4 analytics platform.

The resulting real-time insights enable companies with equipment in challenging operating environments with low bandwidth to develop an optimized proactive maintenance strategy. This leads to reduced downtime and costs, optimized maintenance planning, increased operational efficiency and prolonged machine lifetime. It could also have a positive impact in some industries by lowering risks associated with safety and reducing environmental impact.

HOW IT WORKS

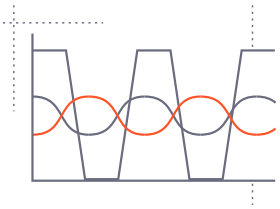
SAM4 Health & Edge Analytics



1

High-quality electrical data acquisition

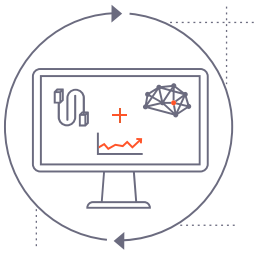
SAM4 uses proprietary IIoT hardware that easily installs inside the motor control cabinet to measure all three phases of current and voltage.



2

Edge analytics

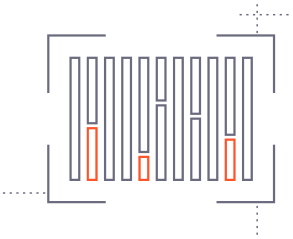
Edge devices perform analytics locally to transform data into a wide variety of failure indicators to greatly reduce data transfer requirements and monitor the asset continuously.



4

AI-powered analytics

SAM4's failure detection models are built with artificial intelligence to make complex detections and improve over time as more failures get processed.



3

Data ingestion to the cloud

Data is encrypted and sent to the Samotics cloud over 4G, Wi-Fi or Ethernet.

5

System-agnostic insights

Failure predictions and asset performance insights are shared via incident notifications, our interactive SAM4 dashboard and a set of APIs to integrate with a system of choice (e.g. CMMS).

Reap the benefits of Edge Analytics

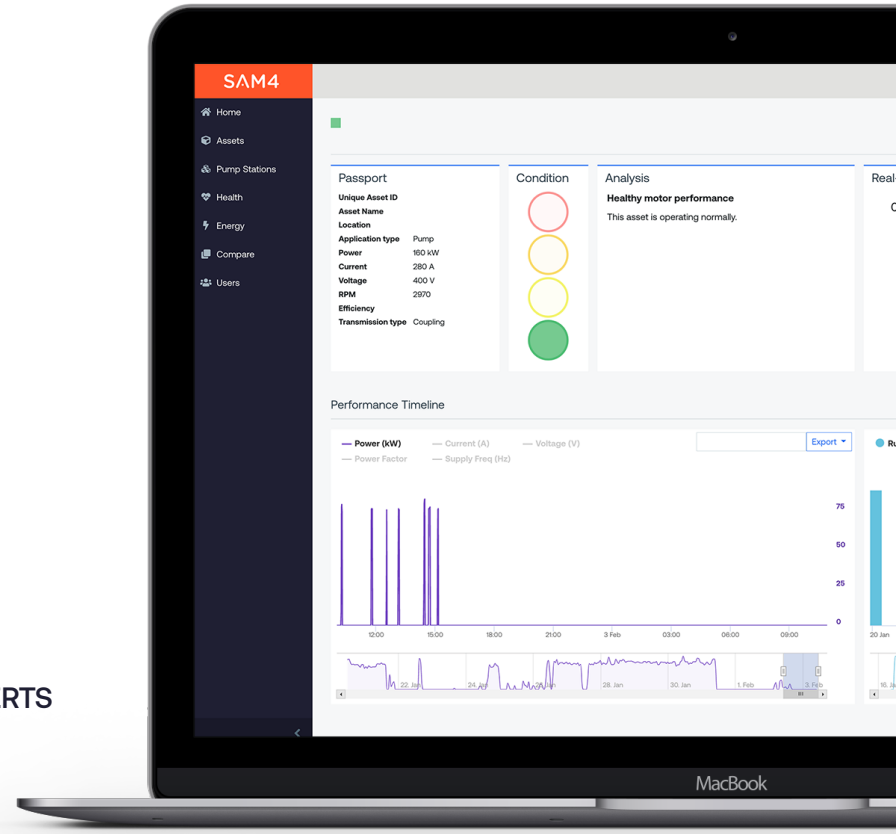
Edge analytics enable SAM4 Health asset monitoring system to perform in places with poor and restrictive network. Companies in low-bandwidth and secure settings can now benefit from a system that offers superior fault detection accuracy with over 90% of failures, both mechanical and electrical, detected up to 5 months in advance. This allows for maintenance scheduling to be optimized with respect to asset repair and replacement. With equipment located in challenging operating environments, customers also benefit from continuous and remote monitoring with sensors installed in the motor control cabinet, not on the equipment itself.

More improvements on the way with SAM4 Health and Edge Analytics

SAM4 Health is continuously improving due to its machine learning algorithms and developments made by Samotics data scientists and engineers. This means that Samotics' customers are always in possession of the latest updates to SAM4 Health, which are performed seamlessly without impacting customers' day-to-day operations. For our customers operating in remote locations there is further good news on the horizon. We continue to reduce our data transferred to the cloud, while at the same time generating higher quality data for better detections and insights. We will achieve this by bringing the intelligence to the Edge and by being more selective in what we send and don't send to the platform. In any instance, insights and detections will always continue to be shared with customers on site.

AI-based asset health monitoring system for your critical equipment

- 24/7, REMOTE CONDITION MONITORING
- MARKET-LEADING DETECTION ACCURACY
- REAL-TIME PUMP PERFORMANCE DATA
- REMOTE, SAFE AND RELIABLE INSTALLATION
- AUTOMATED ANALYSIS AND ALERTS



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